

University of Central Punjab
Faculty of Information Technology and Computer Science

Course Title: Software Construction & Development
Course Code: SESD3242

Assignment 3

Name	ID	Contribution in the assignment (Specifically mention the part which you have done)
Farrukh Shahid	L1F23BSSE0336	I have completed the entirety of this assignment independently.

CLO	At the end of the course the students will be able to:	BT Level*	PLO
CLO2	Apply construction techniques to write a clean and maintainable code that meets industry standards and best practices.	C2	2

Deadline: 24-May-26

Total Marks= 50

Instructions:

1. **Late submission will cause 20% deduction. No submission after 2 Days.**
2. This assignment is a group assignment. 2 members in each group. Individual submissions will not be accepted.
3. **ASSIGNMENT SHOULD BE PROPERLY TYPED.**
4. **Attach this title page to your submission. (5 marks)**
5. **Upload it on the portal on 24-May-26_at 11:55pm**
6. **Plagiarism is seriously your own choice.**
7. **Evaluation will be viva-based. Viva date and time will be announced later.**

Link:

<https://github.com/farrukhshahid25/LibraryBookIssueSytem.git>

List of Implemented Exceptions:

Custom Exceptions:

1. **EmptyFieldException:** Thrown when any required text field (like Student Name, Roll Number, Book Title, or Dates) is left blank.
2. **InvalidRollNumberException:** Thrown when the user inputs letters or special characters in the Roll Number field instead of digits.
3. **NullSelectionException:** Thrown when the user forgets to select a Book Category from the dropdown menu or doesn't select a Book Type (New/Old Edition).
4. **InvalidDateException:** Thrown when the date is not in the correct dd/MM/yyyy format, or if the Return Date is logically earlier than the Issue Date.

Built-in Java Exceptions:

5. **NumberFormatException:** Handled when the system tries to convert string inputs (like dates or roll numbers) into integers using Integer.parseInt(), and the input contains invalid characters.
6. **Exception (General Catch-all):** A generic exception handler added at the end to safely catch any other unexpected run-time errors that might crash the program.

GitBash:

```
G L B@DESKTOP-RPH5MGK MINGW64 ~/Desktop/SCD Assignment #03 (master)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 6 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 312.14 KiB | 9.18 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/farrukhshahid25/LibraryBookIssueSystem.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

G L B@DESKTOP-RPH5MGK MINGW64 ~/Desktop/SCD Assignment #03 (master)
$ |
```

Screenshots:

Library Book Issue System

Library Book Issue System

Student Name	
Roll Number	L1F23BSSE0336
Book Title	Operating Systems
Book Category	
Book Type	
Issue Date	
Return Date	7-24-2026
Remarks	Okay

Error

⚠ Please fill all required fields!

OK

Library Book Issue System

Library Book Issue System

Student Name	Farrukh Shahid
Roll Number	ABCD
Book Title	DSA
Book Category	
Book Type	
Issue Date	
Return Date	25-05-2026
Remarks	Okay

Issue Book

Reset

Exit

Error

Roll Number should contain only digits!

OK

Library Book Issue System

Library Book Issue System

Student Name	Farrukh
Roll Number	0336
Book Title	DSA
Book Category	Error
Book Type	
Issue Date	
Return Date	27-05-2026
Remarks	Okay

Issue Book

Reset

Exit

Error

 Please select a Book Category!

OK

Library Book Issue System

Library Book Issue System

Student Name

Farrukh

Roll Number

0336

Book Title

DSA

Book Category

Book Type

Issue Date

Return Date

21-05-2026

Remarks

Okay

Issue Book

Reset

Exit

Error

Date format should be dd/MM/yyyy!

OK

Library Book Issue System

Library Book Issue System

Student Name

Farrukh

Roll Number

0336

Book Title

DSA

Book Category

Info

Book Type

Issue Date

Return Date

27-05-2026

Remarks

Okay

Operation Completed

OK

Issue Book

Reset

Exit